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Research Paper

The Impact of Scoring on Team Wins

In all sports around the world, every team has its star: the hero that leads their team to victory through thick and thin. However, the conceived notion of the better star allowing a team to win more games may or may not be true. I want to test this theory, specifically through my favorite sport, basketball. In this research project, I am trying to determine how the leading scorer on each NBA team of the 2021-2022 season impacts their teams, and how much they contribute to winning games. I will be considering the top scorer on each NBA the "star" of their team. This research paper will use concepts statistics to determine how much the "star" of each NBA team really impacts the success of their team, and if removing the said "superstar" from each team will impact their winning success, or if their win percentage will remain the same.

In order to determine this answer, I had to statistically determine how the top scorer on each team impacted their team's wins. I first gathered information on NBA players and team statistics, based on the games they played so far during the 2021-2022 season. After gathering the most recent information based on NBA statistics from December 2nd, 2021. This is after each team roughly played about 20 games, about ¼ of the regular season. The following

datasheet shown below is the information I gathered on each of the 30 team's leading scorers. I obtained this information from the official site NBA.com.

Team	#	PLAYER	GP	MIN	PTS	FGM	FGA	FG%	3PM	3PA	3P%	FTM	FTA	FT%
Brookyln Nets	1	Kevin Durant	20	35.8	28.6	10.4	19.2	53.9	1.9	4.8	38.9	6	7	86.3
Golden State Warriors	2	Stephen Curry	20	34.3	27.8	9.2	20.3	45.2	5.4	13.1	41.2	4.1	4.4	94.3
Milwaukee Bucks	3	Giannis Antetokounmpo	21	32.7	27.6	9.8	18.4	53.2	1.1	4.1	27.6	6.9	10.2	67.3
Atlanta Hawks	4	Trae Young	22	34	26.3	9.3	19.9	46.6	2.7	6.9	38.8	5	5.6	89.5
Chicago Bulls	5	DeMar DeRozan	22	35	25.9	9.2	18.8	48.9	0.9	2.6	32.8	6.7	7.5	89.6
Los Angeles Clippers	6	Paul George	21	35.5	25.9	9.5	22	43.1	2.9	8.8	33.2	4	4.5	89.5
Denver Nuggets	7	Nikola Jokic	16	32.7	25.7	10	17.3	58	1.6	4.3	38.2	4.1	5.4	75.6
Dallas Mavericks	8	Luka Doncic	17	34.7	25.6	9.6	21.3	45	2.8	8.2	34.3	3.6	5.5	66.7
Boston Celtics	10	Jayson Tatum	22	36.6	24.4	8.6	21.7	39.7	2.7	8.4	31.9	4.4	5.5	80.8
Los Angeles Lakers	11	Anthony Davis	22	35.7	24.3	9.7	18.7	51.8	0.4	2.1	19.6	4.5	6.2	72.8
Memphis Grizzlies	12	Ja Morant	19	32.6	24.1	8.9	18.8	47.6	1.6	4.6	35.6	4.6	5.9	77.7
Minnesota Timberwolves	13	Karl-Anthony Towns	22	34.3	24	8.5	16.8	50.8	2.6	5.9	44.2	4.3	5.2	81.7
Miami Heat	14	Jimmy Butler	17	34.2	23.6	7.9	15.3	51.9	0.4	1.8	20	7.4	8.6	85
Phoenix Suns	15	Devin Booker	21	32.2	23.2	8.6	18.7	45.8	2.4	5.9	40.3	3.7	4.3	85.7
Utah Jazz	16	Donovan Mitchell	20	32.2	23.2	8.7	20.1	43.4	2.9	9	32.2	2.9	3.3	87.9
Washington Wizards	17	Bradley Beal	19	36.1	22.7	8.7	20.1	43.5	1.7	6.3	26.9	3.6	4	89.5
New Orleans Pelicans	19	Brandon Ingram	17	34.3	21.8	8.2	18.6	44.2	1.8	4.6	38	3.6	4.2	84.7
Portland Trail Blazers	21	Damian Lillard	20	35.7	21.5	7.3	18.4	39.7	2.8	9.1	30.2	4.1	4.5	91.1
Oklahoma City Thunder	22	Shai Gilgeous-Alexander	19	34.9	21.2	7.4	17.9	41.1	2.1	6.8	30	4.4	5.6	78.5
Indiana Pacers	23	Malcolm Brogdon	19	35.4	21	7.5	16.4	46	2	6.2	32.5	3.9	4.5	87.2
Sacramento Kings	27	De'Aaron Fox	23	34.8	20.3	7.5	17.3	43.2	1.2	4.3	27.3	4.1	5.5	74.8
Charlotte Hornets	28	LaMelo Ball	24	33.3	20	7.3	17.3	41.8	2.9	7.5	39.1	2.6	2.9	90
Orlando Magic	29	Cole Anthony	17	34.2	19.9	6.9	16.4	42.1	2.7	7.5	35.9	3.4	3.9	86.6
Toronto Raptors	31	Fred VanVleet	21	38.3	19.9	7.1	16.2	44.1	3.1	8	38.7	2.5	2.9	85.2
Detroit Pistons	33	Jerami Grant	20	32.8	19.5	6.4	15.9	40.4	1.8	5.6	32.1	4.9	5.8	83.6
New York Knicks	34	Julius Randle	21	35.2	19.4	7	16.7	41.7	1.8	5.4	32.5	3.8	5	76
San Antonio Spurs	35	Dejounte Murray	19	34.7	19.1	8.1	18.1	44.6	1.2	3.8	30.6	1.8	2.6	70
Cleveland Cavaliers	42	Darius Garland	20	34.7	18.5	7	15.3	45.8	2.6	7.1	35.9	2	2.3	84.8
Philadelphia 76ers	44	Tyrese Maxey	22	35.6	17.5	6.5	14	46.8	1.2	3.2	36.6	3.2	3.6	88.8
Houston Rockets	59	Christian Wood	21	31.6	16	6.1	13.1	46.4	1.5	4.5	34	2.3	4	57.1

Table 1: 2021-2022 Stats of Top Scorer of Each NBA Team

The number of each player shown is the ranking of the player's PPG vs every other player in the league. After obtaining the PPG, field goal stats, and minutes played, I then gathered information on each NBA teams' winning percentage and total wins. I have the data of these statistics shown below.

Team	TEAM WIN%	TEAM WIN	TEAM PTS
Brookyln Nets	0.714	15	109.6
Golden State Warriors	0.857	18	113.1
Milwaukee Bucks	0.636	14	110.6
Atlanta Hawks	0.545	12	110.5
Chicago Bulls	0.636	14	109.1
Los Angeles Clippers	0.5	11	105.6
Denver Nuggets	0.476	10	103.6
Dallas Mavericks	0.55	11	106
Boston Celtics	0.545	12	106.3
Los Angeles Lakers	0.522	12	111.6
Memphis Grizzlies	0.524	11	110
Minnesota Timberwolves	0.5	11	108
Miami Heat	0.591	13	107.7
Phoenix Suns	0.857	18	112.2
Utah Jazz	0.667	14	113.1
Washington Wizards	0.636	14	106.2
New Orleans Pelicans	0.25	6	103
Portland Trail Blazers	0.5	11	110.4
Oklahoma City Thunder	0.286	6	99
Indiana Pacers	0.375	9	107.4
Sacramento Kings	0.391	9	110
Charlotte Hornets	0.542	13	114.9
Orlando Magic	0.217	5	100
Toronto Raptors	0.409	9	105.2
Detroit Pistons	0.19	4	98.7
New York Knicks	0.524	11	106.5
San Antonio Spurs	0.316	6	107.2
Cleveland Cavaliers	0.545	12	103.9
Philadelphia 76ers	0.5	11	106.5
Houston Rockets	0.238	5	104

Table 2: 2021-2022 Stats of Each NBA Team's Wins and Team PPG

After obtaining the required information, I could then figure out how to make an estimated equation. This would allow me to determine the total amount of wins a team will get based on their top scorer's PPG in a season.

When determining my independent and dependent variables, I initially used percentages for both values. I used the ratio of player PPG by their teams PPG to determine the percentage of points that they contributed to their team and used this as one of my independent variables, x_1. I then used the players FG%, 3P%, and FT% as my other FG%, as x_2, x_3, and x_4 respectively. I

used TEAM WIN% as my dependent variable, y, since we are trying to determine how team winning success is altered based on the scoring of the top scorer of the team. After setting up the data on a chart, and determining its multiple regression through excel, here is the data I collected. Keep in mind that I had to divide the stated FG%, 3P%, and FT% by 100 to keep the decimal values true to the TEAM WIN %.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.583167
R Square	0.3400837
Adjusted R S	0.2344971
Standard Err	0.1478056
Observations	30

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	0.2814616	0.0703654	3.2208984	0.0290975
Residual	25	0.5461627	0.0218465		
Total	29	0.8276243			

	Coefficients	*tandard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-0.879614	0.4419885	-1.990129	0.0576185	-1.789906	0.0306784	-1.789906	0.0306784
PTS%	1.8564837	1.15985	1.6006239	0.1220227	-0.532272	4.2452395	-0.532272	4.2452395
FG%	0.9195835	0.7340637	1.2527298	0.2218958	-0.592249	2.431416	-0.592249	2.431416
3P%	0.0151386	0.5043164	0.0300181	0.976291	-1.02352	1.0537976	-1.02352	1.0537976
FT%	0.6871242	0.3542107	1.9398743	0.0637563	-0.042386	1.4166347	-0.042386	1.4166347

Figure 1: Multiple Regression of PTS%, FG%, 3P%, & FT% vs TEAM WIN %

Estimated Equation: $\hat{y}= -0.88 + 1.8x_1 + 0.92x_2 + 0.015x_3 + 0.69x_4$

After solving the multiple regression, and obtaining the following data, I was able to get an estimated equation for the four independent variables. The equation came out as $\hat{y}= -0.88 + 1.8x_1 + 0.92x_2 + 0.015x_3 + 0.69x_4$. This is the first multiple regression possibility that I ran.

However, I did not like the independent variables I used for this regression, because I believed it used too many variables, and that the variables overlapped each other. Also, the R^2 value was very low, at 0.34, so I realized that the variables didn't correlate with each other enough to make a solid regression. Taking this into consideration, I rearranged the independent variables. For my second, more improved multiple regression, I kept the same dependent variable, TEAM WIN %, but altered the independent variables to just PLAYER PPG as x_1, and TEAM PPG as x_2. This allowed me to focus my independent variables on the average points the top scorer on each team scores per game to the average points the team scores, and then directly compare this to the team's winning success. Therefore, I obtain a more direct comparison the player's impact. After determining the new multiple regression using the new independent variables, I was able to formulate a new estimated equation for TEAM WIN %. Graphs and data tables surrounding this regression is shown below.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.817308947
R Square	0.667993915
Adjusted R S	0.643400871
Standard Er	10.08806233
Observation	30

ANOVA

	df	*SS*	*MS*	*F*	*ignificance F*
Regression	2	5528.48	2764.24	27.161905	3.431E-07
Residual	27	2747.763	101.769		
Total	29	8276.243			

	Coefficients	*tandard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-249.4620623	49.72673	-5.016659	2.912E-05	-351.4929	-147.4312	-351.4929	-147.4312
PTS	2.083090797	0.6301293	3.3058149	0.002681	0.7901723	3.3760093	0.7901723	3.3760093
TEAM PTS	2.35236646	0.5000951	4.7038381	6.751E-05	1.326256	3.3784769	1.326256	3.3784769

RESIDUAL OUTPUT

Observation	Predicted TEAM W	Residuals
1	67.93369847	3.46630153
2	74.50050844	11.1994916
3	68.20297413	-4.6029741
4	65.25971945	-10.759719
5	61.13317008	2.46682992
6	52.89988747	-2.8998875
7	47.77853639	-0.1785364
8	53.21590682	1.78409318
9	51.4219078	3.0780922
10	63.68114096	-11.481141
11	59.50073646	-7.1007365
12	54.58769446	-4.5876945
13	53.04874821	6.05125179
14	62.80116096	22.898839
15	64.91829077	1.78170923
16	47.6454168	15.9545832
17	38.24306241	-13.243062
18	55.02564698	-5.025647
19	27.58374209	1.01625791
20	46.9270022	-9.4270022
21	51.58499144	-12.484991
22	62.48665985	-8.2866599
23	27.22809052	-5.5280905
24	39.46039611	1.43960389
25	23.3367778	-4.3367778
26	41.47692711	10.9230729
27	42.49865639	-10.898656
28	33.48599259	21.0140074
29	37.51905459	12.4809454
30	28.51350225	-4.7135022

Figures 2 & 3: Multiple Regression +Residual Output of PPG & TEAM PPG vs TEAM WIN%

New Estimated Equation: $\hat{y}= -249.462 + 2.083x_1 + 2.352x_2$

After determining the new multiple regression model, I was able to make a new estimated equation for the dependent variable, y. The following equation came out to be $\hat{y}= -249.462 + 2.083x_1 + 2.352x_2$. The new independent variables of the new regression model are more solid, as the comparison of player PPG vs TEAM PPG helps show the player's impact on team win success. This is proved by the higher R^2 value, as it increased from 0.34 to 0.67. The proof is also shown through the residual correlation between each dependent and independent variable. Plots for this comparison are shown below.

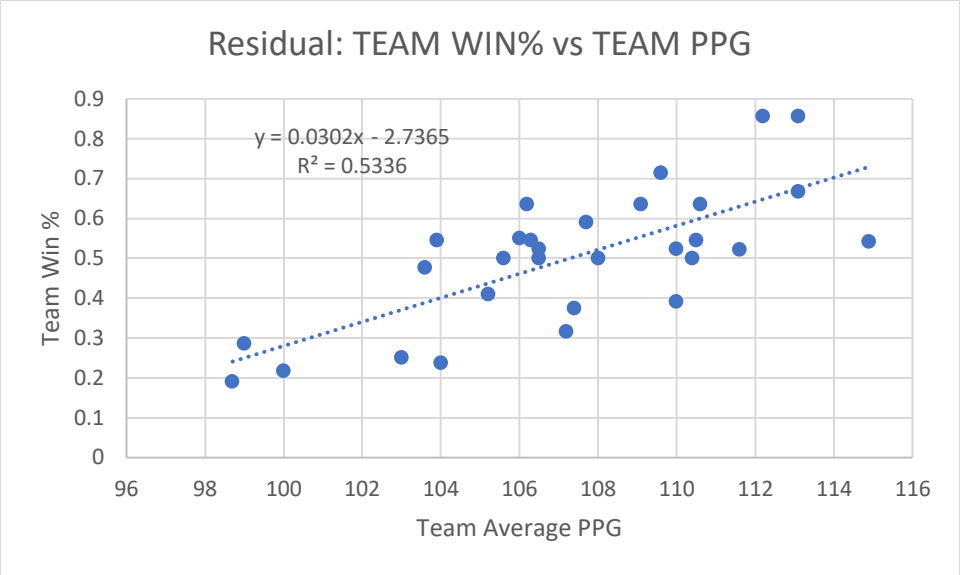


Figure 4: Residual Plot: Team Win% vs Team PPG

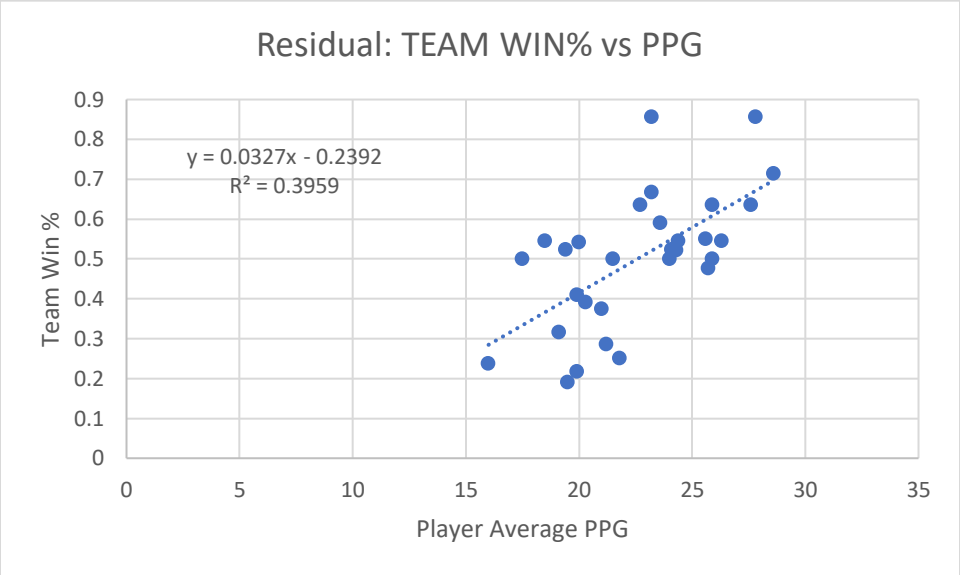


Figure 4: Residual Plot: Team Win% vs Player PPG

From the plots shown above, we can see that there is a positive linear relationship between x_1 and y, and between x_2 and y. There is a moderate error, due to a small sample size of 30, but the error is far greater for the PLAYER PPG vs TEAM PPG. Although this plot

simplifies that a higher PPG by the team or player will result in a higher TEAM WIN %, many variances in the player residual plot indicate that this may not always be the case. For example, Kevin Durant has a higher PPG than Devin Booker, 28.6 to 23.2, yet the Brooklyn Nets have a lower win percentage than the Phoenix Suns, 71.4% to 85.7%. This proves that the top scorer in the league may not always relate to the best team's win percentage. This is why multiple regression must be made: to determine a better evaluation. Even with the preconceived notion of more PPGs leading to a higher WIN%, there is variance shown through the residual output, shown in Figure 4. The figure predicts that teams such as the Los Angeles Lakers (10th observation) and Utah Jazz (15th) will have a higher WIN% than a team such as the Los Angeles Clippers (6th observation) or Denver Nuggets (7th observation), although Paul George and Nikola Jokic has a higher PPG average versus Anthony Davis and Donovan Mitchell.

After determining the new estimate equation, $\hat{y}= -249.462 + 2.083x_1 + 2.352x_2$, and determining the new predicted values for each NBA team's win percentage, this is what I can conclude. The top scorer based on one NBA team does tend to have a higher win percentage than the top scorer of another team if the former player has a higher PPG than the latter. However, this is not necessarily always the case, as explained in the previous paragraph, which proves that being the most talented scorer in the league will give you one of the best win percentages in the league, but will not guarantee THE best win percentage. However, with the following estimate equation, the win percentage based on an NBA top scorer's impact can be calculated to a more accurate degree, to avoid the consensus that the team with the better scorer will automatically have a higher win percentage.

Works Cited

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